

5.1 Ажлын зорилго

Энэхүү лабораторийн ажлаар бид IP датаграм буюу сүлжээний давхаргад ашигладаг IP протоколын толгойн хэсгийн бүтцийг судална. Мөн traceroute програмын үр дүнд үүсэх илгээгч, хүлээн авагч IP datagram-ийг анализ хийнэ. Лабораторийн ажлын төгсгөлд IP datagram-ийн талбарууд, IP фрагментийн талаар мэдлэгтэй болно.

5.2 Үндсэн ойлголт

IP протокол нь дамжуулагчаас хүлээн авагчруу датаграмыг хүргэх үүрэгтэй. Өөрөөр хэлбэл IP хаягтай ажиллаж чадах төхөөрөмжүүдийн буюу хостуудыг логик холбоосоор холбож пакет дамжуулах боломжтой болгоно. Сүлжээгээр дамжуулагдаж байгаа пакетыг толгой хэсгийг задлаж үзвэл IP-тай холбоотой талбарууд байх бөгөөд илгээгч (source), хүлээн авагч (destination) гэсэн IP хаяг агуулах талбаруудад үндэслэн сүлжээний төхөөрөмжүүд үйлдэл гүйцэтгэдэг.

Илгээгч болон хүлээн авагч эцсийн төхөөрөмжүүд (хостууд) -ийн үүсгэсэн пакет нь ашиглаж буй протокол, сүлжээ, төхөөрөмжийн онцлогоос хамаарч хэмжээ нь өөрчлөгдөж байдаг. Гэтэл сүлжээний дамжуулах төхөөрөмжүүдийн хүчин чадал, дамжууллын орчны зурвасын өргөн зэргээс хамаарч том хэмжээтэй пакетыг илгээх боломжгүй болох үед боломжтой хэмжээгээр хувааж дамжуулна. Энэ үйлдлийг бид fragmentation буюу хуваагдал гэх бөгөөд хуваагдсан пакетуудыг эцсийн хүлээн авагч хүлээж аван нэгтгэдэг байна.

IP нь алдааны удирдлагагүй мөн тусламжийн механизм байхгүй хоёр дутагдалтай талыг сайжруулах зорилгоор Internet Control Message Protocol (ICMP) загварчилж гаргасан.

Бид энэхүү лабораторийн ажлаар traceroute (tracert) болон ping зэрэг командуудыг ашиглаж өөрийн төхөөрөмжөөс сонгосон сервер рүү пакетууд үүсгэж илгээнэ. Илгээхээс гадна хариу хүлээн авах процессыг Wireshark программаар бичиж авч ажиглалт хийнэ.

5.3 Туршилт

Wireshark программаа ажлуулж дамжуулж байгаа пакетуудыг бичиж авах үйлдлийг идэвхжүүл.

1. Командын мөр (cmd) дээр ping команд ашиглаж дурын сервер болон өөрийн холбогдсон байгаа сүлжээний гарц руу мэдээлэл дамжуулаарай. Ингэхдээ дараах байдлаар хэд хэдэн удаа гүйцэтгэ.

- a. Ping server_name
- b. Ping server_name -l 128
- c. Ping server_name -l 256

- d. Ping server_name -l 512
- e. Ping server_name -l 1024
- f. Ping server_name -l 1536
- g. ping your_gateway -l 2000
- h. ping your_gateway -l 3800

2. Командын мөр (cmd) дээр tracert команд ашиглаж дурын сервер рүү мэдээлэл дамжуул.

3. Цуглуулсан пакетуудад дүн шинжилгээ хийж дараах даалгаваруудыг хийж гүйцэтгэнэ.

```
munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~$ ping -c 4 google.com
ping -c 4 -s 128 google.com
ping -c 4 -s 256 google.com
ping -c 4 -s 512 google.com
ping -c 4 -s 1024 google.com
ping -c 4 -s 1536 google.com
```

```
munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~$ ping -c 4 -s 3800 10.3.200.1
PING 10.3.200.1 (10.3.200.1) 3800(3828) bytes of data.
3808 bytes from 10.3.200.1: icmp_seq=1 ttl=254 time=9.01 ms
3808 bytes from 10.3.200.1: icmp_seq=2 ttl=254 time=13.0 ms
3808 bytes from 10.3.200.1: icmp_seq=3 ttl=254 time=8.77 ms
3808 bytes from 10.3.200.1: icmp_seq=4 ttl=254 time=13.0 ms
```

```
munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~$ traceroute google.com
traceroute to google.com (142.250.71.174), 64 hops max
 1  10.3.200.1  6.908ms  9.694ms  6.717ms
 2  10.9.1.9  6.143ms  110.218ms  5.574ms
 3  10.0.2.9  6.456ms  6.473ms  6.449ms
 4  119.40.100.41  3.048ms  6.385ms  6.399ms
 5  119.40.100.17  107.217ms  6.184ms  3.155ms
 6  180.149.95.209  3.234ms  3.137ms  3.202ms
 7  180.149.92.61  6.497ms  6.498ms  6.292ms
 8  180.149.97.158  111.121ms  191.444ms  194.115ms
 9  * * *
10  * * *
11  10.3.202.234  2951.918ms !H 142.251.244.222  2148.482ms  65.256ms
```

Command lines

5.4 Даалгавар

1. Илгээгчийн IP хаягийн талбарт байгаа хаягийг бич. Server_name -ийн хаяг ямар талбарт байна вэ?

Илгээгчийн IP хаяг нь **IP header-ийн “Source IP address”** талбарт байрлана. Харин server_name буюу хүлээн авагчийн IP хаяг нь **“Destination IP address”** талбарт бичигддэг.

- Source IP: 10.3.202.234
- Destination IP: 142.250.71.174

782	15.747241848	10.3.202.234	142.250.71.174	ICMP	554 Echo (ping) request	id
783	16.069594157	142.250.71.174	10.3.202.234	ICMP	554 Echo (ping) reply	id
784	16.748706539	10.3.202.234	142.250.71.174	ICMP	554 Echo (ping) request	id
785	16.862167293	142.250.71.174	10.3.202.234	ICMP	554 Echo (ping) reply	id
786	16.869485491	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id
787	17.015309240	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id
788	17.871683158	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id
790	18.083929534	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id
798	18.873253583	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id
799	18.989274319	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id
821	19.874681369	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id
854	21.054376853	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id
898	22.078173715	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id
932	23.102250947	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id
2214	57.553161339	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable	
2223	57.553234130	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable	
2226	57.553249984	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable	
2231	57.553265980	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable	
2236	57.553298539	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable	
2239	57.553311416	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable	

▶ Frame 782: 554 bytes on wire (4432 bits), 554 bytes captured (4432 bits) on interface wlo1, id 0
 ▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)
 ▶ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174
 ▶ Internet Control Message Protocol

Source address and destination address

2. Layer protocol field –ийн хэмжээ хэд вэ? Ping командад -l сонголт өгөхөд өөрчлөгдөж байна уу? Үр дүнг тайлбарла.

IP header доторх **Protocol field** нь **8 бит (1 byte)** хэмжээтэй. Ping команд нь **ICMP (Protocol = 1)** ашигладаг. ping -s (packet size) утга өөрчлөгдөхөд **Protocol field** өөрчлөгдөхгүй, харин **Total Length** талбарын утга өөрчлөгдөнө.

708	11.678152067	10.3.202.234	142.250.71.174	ICMP	298	Echo (ping) request	id=0xba60, seq=2/512,
722	12.036935038	142.250.71.174	10.3.202.234	ICMP	298	Echo (ping) reply	id=0xba60, seq=2/512,
723	12.679049075	10.3.202.234	142.250.71.174	ICMP	298	Echo (ping) request	id=0xba60, seq=3/768,
725	12.873767209	142.250.71.174	10.3.202.234	ICMP	298	Echo (ping) reply	id=0xba60, seq=3/768,
729	13.680916363	10.3.202.234	142.250.71.174	ICMP	298	Echo (ping) request	id=0xba60, seq=4/1024,
730	13.739934819	142.250.71.174	10.3.202.234	ICMP	298	Echo (ping) reply	id=0xba60, seq=4/1024,
731	13.746728310	10.3.202.234	142.250.71.174	ICMP	554	Echo (ping) request	id=0xba7a, seq=1/256,
732	14.015832362	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=0xba7a, seq=1/256,
733	14.747027783	10.3.202.234	142.250.71.174	ICMP	554	Echo (ping) request	id=0xba7a, seq=2/512,
756	15.178308292	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=0xba7a, seq=2/512,
782	15.747241848	10.3.202.234	142.250.71.174	ICMP	554	Echo (ping) request	id=0xba7a, seq=3/768,
783	16.069594157	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=0xba7a, seq=3/768,
784	16.748706539	10.3.202.234	142.250.71.174	ICMP	554	Echo (ping) request	id=0xba7a, seq=4/1024,
785	16.862167293	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=0xba7a, seq=4/1024,
786	16.869485491	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=0xba8a, seq=1/256,
787	17.015309240	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=0xba8a, seq=1/256,
788	17.871683158	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=0xba8a, seq=2/512,
790	18.083929534	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=0xba8a, seq=2/512,
798	18.873253583	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=0xba8a, seq=3/768,
799	18.989274319	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=0xba8a, seq=3/768,
▶ Frame 756: 554 bytes on wire (4432 bits), 554 bytes captured (4432 bits) on interface wlo1, id 0 ▶ Ethernet II, Src: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2), Dst: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1) ▶ Internet Protocol Version 4, Src: 142.250.71.174, Dst: 10.3.202.234 0100 = Version: 4 0101 = Header Length: 20 bytes (5) ▶ Differentiated Services Field: 0x60 (DSCP: CS3, ECN: Not-ECT) Total Length: 540 Identification: 0x0000 (0) 0000 = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 113 Protocol: ICMP (1) Header Checksum: 0x9beb [validation disabled] [Header checksum status: Unverified] Source Address: 142.250.71.174 Destination Address: 10.3.202.234 ▶ Internet Control Message Protocol							
	0000	50	5a	65	cc		
	0010	02	1c	00	00		
	0020	ca	ea	00	00		
	0030	00	00	9b	aa		
	0040	16	17	18	19		
	0050	26	27	28	29		
	0060	36	37	38	39		
	0070	46	47	48	49		
	0080	56	57	58	59		
	0090	66	67	68	69		
	00a0	76	77	78	79		
	00b0	86	87	88	89		
	00c0	96	97	98	99		
	00d0	a6	a7	a8	a9		
	00e0	b6	b7	b8	b9		
	00f0	c6	c7	c8	c9		
	0100	d6	d7	d8	d9		

3. IP header –д IP datagram –ийн payload-д хэдэн byte байна? Payload bytes –ийн тоо?

Payload Bytes-ийн Тоо:

- **Payload Length = Total Length - Header Length**
540 bytes – 20 bytes = 520 bytes
- **WiIP Payload нь 128 (ICMP data) +8 (ICMP header) = 136 байт орчим байна.**

708	11.678152067	10.3.202.234	142.250.71.174	ICMP	298 Echo (ping) request	id=0xba60, seq=2/512,
722	12.036935038	142.250.71.174	10.3.202.234	ICMP	298 Echo (ping) reply	id=0xba60, seq=2/512,
723	12.679049075	10.3.202.234	142.250.71.174	ICMP	298 Echo (ping) request	id=0xba60, seq=3/768,
725	12.873767209	142.250.71.174	10.3.202.234	ICMP	298 Echo (ping) reply	id=0xba60, seq=3/768,
729	13.680916363	10.3.202.234	142.250.71.174	ICMP	298 Echo (ping) request	id=0xba60, seq=4/1024,
730	13.739934819	142.250.71.174	10.3.202.234	ICMP	298 Echo (ping) reply	id=0xba60, seq=4/1024,
731	13.746728310	10.3.202.234	142.250.71.174	ICMP	554 Echo (ping) request	id=0xba7a, seq=1/256,
732	14.015832362	142.250.71.174	10.3.202.234	ICMP	554 Echo (ping) reply	id=0xba7a, seq=1/256,
733	14.747027783	10.3.202.234	142.250.71.174	ICMP	554 Echo (ping) request	id=0xba7a, seq=2/512,
756	15.178308292	142.250.71.174	10.3.202.234	ICMP	554 Echo (ping) reply	id=0xba7a, seq=2/512,
782	15.747241848	10.3.202.234	142.250.71.174	ICMP	554 Echo (ping) request	id=0xba7a, seq=3/768,
783	16.069594157	142.250.71.174	10.3.202.234	ICMP	554 Echo (ping) reply	id=0xba7a, seq=3/768,
784	16.748706539	10.3.202.234	142.250.71.174	ICMP	554 Echo (ping) request	id=0xba7a, seq=4/1024,
785	16.862167293	142.250.71.174	10.3.202.234	ICMP	554 Echo (ping) reply	id=0xba7a, seq=4/1024,
786	16.869485491	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id=0xba8a, seq=1/256,
787	17.015309240	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id=0xba8a, seq=1/256,
788	17.871683158	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id=0xba8a, seq=2/512,
790	18.083929534	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id=0xba8a, seq=2/512,
798	18.873253583	10.3.202.234	142.250.71.174	ICMP	1066 Echo (ping) request	id=0xba8a, seq=3/768,
799	18.989274319	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply	id=0xba8a, seq=3/768,
Frame 756: 554 bytes on wire (4432 bits), 554 bytes captured (4432 bits) on interface wlo1, id 0						
Ethernet II, Src: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2), Dst: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1)						
Internet Protocol Version 4, Src: 142.250.71.174, Dst: 10.3.202.234						
0100 = Version: 4						
.... 0101 = Header Length: 20 bytes (5)						
Differentiated Services Field: 0x60 (DSCP: CS3, ECN: Not-ECT)						
Total Length: 540						
Identification: 0x0000 (0)						
0000 = Flags: 0x0						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 113						
Protocol: ICMP (1)						
Header Checksum: 0x9beb [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 142.250.71.174						
Destination Address: 10.3.202.234						
Internet Control Message Protocol						

4. IP datagram fragmented хийгдсэн үү? Яагаад?

Жижиг пакетуудад fragmentation явагдахгүй (ping -s 512, 1024). Харин **packet size > MTU (1500 bytes)** үед fragmentation үүсдэг. Тиймээс ping -s 1536 үед IP datagram fragmentation хийгдсэн байна.

836	20.019724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xbaa0, seq=1/256, ttl=64 (no response found!)
854	21.954376853	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xbaa0, seq=2/512, ttl=64 (no response found!)
898	22.076173715	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xbaa0, seq=3/768, ttl=64 (no response found!)
932	23.102250947	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xbaa0, seq=4/1024, ttl=64 (no response found!)
2214	57.553161339	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable (Port unreachable)	
2223	57.553234130	10.3.202.234	142.250.197.227	ICMP	590 Destination unreachable (Port unreachable)	
Frame 898: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlo1, id 0						
Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)						
Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174						
0100 = Version: 4						
.... 0101 = Header Length: 20 bytes (5)						
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 84						
Identification: 0x3628 (13864)						
0000 = Flags: 0x0						
...0 0000 1011 1001 = Fragment Offset: 1480						
Time to Live: 64						
Protocol: ICMP (1)						
Header Checksum: 0x9832 [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 10.3.202.234						
Destination Address: 142.250.71.174						
[2 IPv4 Fragments (1544 bytes): #897(1480), #898(64)]						
[Frame: 897, payload: 0-1479 (1480 bytes)]						
[Frame: 898, payload: 1480-1543 (64 bytes)]						
[Fragment count: 2]						
[Reassembled IPv4 length: 1544]						
[Reassembled IPv4 data [truncated]: 0800cfaabaa000036694f86880000000c375020000000000101121314..						
Internet Control Message Protocol						

5. Нэг удаа ping команд ажиллуулахад хэдэн ICMP датаграм илгээж байна вэ? Тэдгээр нь хоорондоо ялгаатай юу?

ping -с 4 команд нь **4 ICMP Echo Request** илгээнэ. Тэдгээрийн **Identification, Sequence number, Checksum**, мөн **Timestamp** утгууд өөр байна. Харин толгойн бүтэц, Protocol = 1 зэрэг нь ижил.

Компьютерын сүлжээ (ECEN213)

```

576 6.256760114 10.3.202.234 142.250.71.174 ICMP 98 Echo (ping) request id=0xba3b, seq=3/768, ttl=64 (reply in 577)
577 6.432264271 142.250.71.174 10.3.202.234 ICMP 98 Echo (ping) reply id=0xba3b, seq=3/768, ttl=113 (request in 576)
631 7.252789613 10.3.202.234 142.250.71.174 ICMP 98 Echo (ping) request id=0xba3b, seq=4/1024, ttl=64 (reply in 632)
632 7.372998283 142.250.71.174 10.3.202.234 ICMP 98 Echo (ping) reply id=0xba3b, seq=4/1024, ttl=113 (request in 631)
633 7.377758300 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=1/256, ttl=64 (reply in 642)
642 7.523183137 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) reply id=0xba53, seq=1/256, ttl=113 (request in 633)
659 8.379359288 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=2/512, ttl=64 (reply in 660)
660 8.666146810 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) reply id=0xba53, seq=2/512, ttl=113 (request in 659)
667 9.380386050 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=3/768, ttl=64 (reply in 668)
668 9.436884194 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) reply id=0xba53, seq=3/768, ttl=113 (request in 667)
689 10.382155717 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=4/1024, ttl=64 (reply in 690)
690 10.672070240 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) reply id=0xba53, seq=4/1024, ttl=113 (request in 689)
691 10.676686946 10.3.202.234 142.250.71.174 ICMP 298 Echo (ping) request id=0xba60, seq=1/256, ttl=64 (reply in 692)
692 10.975997708 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) reply id=0xba60, seq=1/256, ttl=113 (request in 691)
708 11.678152967 10.3.202.234 142.250.71.174 ICMP 298 Echo (ping) request id=0xba60, seq=2/512, ttl=64 (reply in 722)

```

6. Ping командуудын эхний датаграмуудыг хооронд нь харьцуул. Ялгаа байна уу? Хоёр дахь датаграмууд ялгаатай байна уу?

Эхний болон дараагийн датаграмуудын хооронд ялгаа нь:

- **Total Length, Payload size, Checksum, Identification, Sequence number** өөрчлөгддөг.
- **Source/Destination IP, Protocol, Version, Header Length** зэрэг нь тогтмол. Өөрөөр хэлбэл, payload-ийн хэмжээ болон тодорхой хяналтын утгууд л ялгаатай байна.

632 7.372998283 142.250.71.174 10.3.202.234 ICMP 98 Echo (ping) request id=0xba3b, seq=3/768, ttl=64 (reply in 577)	632 7.372998283 142.250.71.174 10.3.202.234 ICMP 98 Echo (ping) request id=0xba3b, seq=3/768, ttl=64 (reply in 577)
633 7.377758300 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) reply id=0xba53, seq=1/256, ttl=64 (reply in 642)	633 7.377758300 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) reply id=0xba53, seq=1/256, ttl=64 (reply in 642)
642 7.523183137 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=2/512, ttl=64 (reply in 660)	642 7.523183137 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=2/512, ttl=64 (reply in 660)
659 8.379359288 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=3/768, ttl=64 (reply in 668)	659 8.379359288 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=3/768, ttl=64 (reply in 668)
660 8.666146810 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=4/1024, ttl=64 (reply in 690)	660 8.666146810 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=4/1024, ttl=64 (reply in 690)
667 9.380386050 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=5/1024, ttl=64 (reply in 691)	667 9.380386050 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=5/1024, ttl=64 (reply in 691)
668 9.436884194 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=6/1024, ttl=64 (reply in 692)	668 9.436884194 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=6/1024, ttl=64 (reply in 692)
689 10.382155717 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=7/1024, ttl=64 (reply in 693)	689 10.382155717 10.3.202.234 142.250.71.174 ICMP 170 Echo (ping) request id=0xba53, seq=7/1024, ttl=64 (reply in 693)
690 10.672070240 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=8/1024, ttl=64 (reply in 694)	690 10.672070240 142.250.71.174 10.3.202.234 ICMP 170 Echo (ping) request id=0xba53, seq=8/1024, ttl=64 (reply in 694)
691 10.676686946 10.3.202.234 142.250.71.174 ICMP 298 Echo (ping) request id=0xba60, seq=1/256, ttl=64 (reply in 692)	691 10.676686946 10.3.202.234 142.250.71.174 ICMP 298 Echo (ping) request id=0xba60, seq=1/256, ttl=64 (reply in 692)
692 10.975997708 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=2/512, ttl=64 (reply in 693)	692 10.975997708 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=2/512, ttl=64 (reply in 693)
708 11.678152967 10.3.202.234 142.250.71.174 ICMP 298 Echo (ping) request id=0xba60, seq=3/512, ttl=64 (reply in 694)	708 11.678152967 10.3.202.234 142.250.71.174 ICMP 298 Echo (ping) request id=0xba60, seq=3/512, ttl=64 (reply in 694)
722 12.036935038 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=4/512, ttl=64 (reply in 695)	722 12.036935038 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=4/512, ttl=64 (reply in 695)
723 12.679049075 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=5/512, ttl=64 (reply in 696)	723 12.679049075 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=5/512, ttl=64 (reply in 696)
725 12.873767209 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=6/512, ttl=64 (reply in 697)	725 12.873767209 142.250.71.174 10.3.202.234 ICMP 298 Echo (ping) request id=0xba60, seq=6/512, ttl=64 (reply in 697)

7. Бүх датаграмуудыг ажигла. Толгой хэсгийн аль талбар тогтмол, мөн өөрчлөгдөж байна вэ? Яагаад?

IP Толгойн Талбар	Тогтмол/Өөрчлөгдөж буй	Тайбар
Version	Тогтмол (4 - IPv4)	Ашиглаж буй IP-ийн хувилбар.
Header Length (IHL)	Тогтмол (5 word буюу 20 byte)	Сонголт байхгүй үед толгойн хэмжээ тогтмол байна.
Source Address	Тогтмол	Таны компьютерийн IP хаяг.
Destination Address	Тогтмол	Server_name-ийн IP хаяг.
Protocol	Тогтмол (1)	ICMP-ийн дугаар.
Identification	Өөрчлөгдөж буй	Шифрлэсэн байдлаар нэг ping-ийн датаграмууд бүр нэмэгдэж, ping команд бүр

Компьютерын сүлжээ (ECEN213)

		шинээр эхэлж болно. Датаграмыг нэгтгэхэд ашигладаг.
Total Length	Өөрчлөгдөж буй	–1 сонголтоос хамаарч өөрчлөгдөнө.
Time to Live (TTL)	Өөрчлөгдөж буй	Эхлэлийн TTL тогтмол боловч, дамжуулагч төхөөрөмж (router) бүр 1-ээр хасагддаг.
Header Checksum	Өөрчлөгдөж буй	Толгой хэсгийн утгууд өөрчлөгдөх (жишээ нь, TTL буурах) бүрт дахин тооцоологддог.

516 4.247926083	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xba3b, seq=1/256, ttl=64 (reply in 517)
517 4.431858431	142.250.71.174	10.3.202.234	ICMP	98 Echo (ping) reply	id=0xba3b, seq=1/256, ttl=113 (request in 516)
559 5.249112588	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xba3b, seq=2/512, ttl=64 (reply in 561)
561 5.484493726	142.250.71.174	10.3.202.234	ICMP	98 Echo (ping) reply	id=0xba3b, seq=2/512, ttl=113 (request in 559)
570 6.290709814	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xba3b, seq=3/768, ttl=64 (reply in 577)
577 6.432204271	142.250.71.174	10.3.202.234	ICMP	98 Echo (ping) reply	id=0xba3b, seq=3/768, ttl=113 (request in 570)
631 7.252709613	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request	id=0xba3b, seq=4/1024, ttl=64 (reply in 632)
632 7.372986283	142.250.71.174	10.3.202.234	ICMP	98 Echo (ping) reply	id=0xba3b, seq=4/1024, ttl=113 (request in 631)
633 7.377750380	10.3.202.234	142.250.71.174	ICMP	170 Echo (ping) request	id=0xba53, seq=1/256, ttl=64 (reply in 642)
642 7.523183137	142.250.71.174	10.3.202.234	ICMP	170 Echo (ping) reply	id=0xba53, seq=1/256, ttl=113 (request in 633)
659 8.379359288	10.3.202.234	142.250.71.174	ICMP	170 Echo (ping) request	id=0xba53, seq=2/512, ttl=64 (reply in 669)
660 8.405146103	142.250.71.174	10.3.202.234	ICMP	170 Echo (ping) reply	id=0xba53, seq=2/512, ttl=113 (request in 659)
667 9.389366556	10.3.202.234	142.250.71.174	ICMP	170 Echo (ping) request	id=0xba53, seq=3/768, ttl=64 (reply in 668)
668 9.436804194	142.250.71.174	10.3.202.234	ICMP	170 Echo (ping) reply	id=0xba53, seq=3/768, ttl=113 (request in 667)
689 10.382155717	10.3.202.234	142.250.71.174	ICMP	170 Echo (ping) request	id=0xba53, seq=4/1024, ttl=64 (reply in 690)
690 10.672870240	142.250.71.174	10.3.202.234	ICMP	170 Echo (ping) reply	id=0xba53, seq=4/1024, ttl=113 (request in 689)
691 10.676686946	10.3.202.234	142.250.71.174	ICMP	298 Echo (ping) request	id=0xba60, seq=1/256, ttl=64 (reply in 692)
692 10.975997708	142.250.71.174	10.3.202.234	ICMP	298 Echo (ping) reply	id=0xba60, seq=1/256, ttl=113 (request in 691)
708 11.078152967	10.3.202.234	142.250.71.174	ICMP	298 Echo (ping) request	id=0xba60, seq=2/512, ttl=64 (reply in 722)
722 12.030035038	142.250.71.174	10.3.202.234	ICMP	298 Echo (ping) reply	id=0xba60, seq=2/512, ttl=113 (request in 708)
723 12.079049075	10.3.202.234	142.250.71.174	ICMP	298 Echo (ping) request	id=0xba60, seq=3/768, ttl=64 (reply in 725)
725 12.873767209	142.250.71.174	10.3.202.234	ICMP	298 Echo (ping) reply	id=0xba60, seq=3/768, ttl=113 (request in 723)

8. Identification ба TTL талбарын утгууд хэд байна вэ? Ping командууд хооронд эдгээр утгууд ялгаатай байна уу, тайлбарла.

- **Identification** талбар: Пакет бүрд өвөрмөц утга авдаг (жишээ нь 0x44f6). Энэ нь хуваагдсан пакетуудыг нэгтгэхэд ашиглагдана.
- **TTL** утга: Linux системд ихэвчлэн 64 байна.
→ Ping бүрийн үед TTL эхлэл утга ижил боловч, замын урт (hop count)-аас хамаарч буурч болно.

<pre> 898 22.078173715 10.3.202.234 142.250.71.174 ICMP 98 Echo (ping) request id= 892 23.102268947 10.3.202.234 142.250.71.174 ICMP 98 Echo (ping) request id= 2214 57.553161339 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2223 57.553234130 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2226 57.553249984 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2231 57.553265980 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2236 57.553298539 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2239 57.553311416 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2252 57.559644075 10.3.202.234 142.250.197.196 ICMP 192 Destination unreachable 2405 58.352898342 10.3.202.234 10.3.200.1 ICMP 370 Destination unreachable 2408 58.364703727 10.3.202.234 10.3.200.1 ICMP 370 Destination unreachable 4090 70.933922154 10.3.202.234 10.3.200.1 ICMP 370 Echo (ping) request id= 4099 70.933940389 10.3.202.234 10.3.200.1 ICMP 370 Destination unreachable 4713 77.051972670 10.3.202.234 10.3.202.234 ICMP 562 Echo (ping) reply id= 6293 84.345712579 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6296 84.354041706 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) reply id= 6406 85.347179984 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6410 85.369051853 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= 6677 86.348741703 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6682 86.357349398 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= 6748 87.349929049 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6753 87.362775437 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= </pre>	<pre> 898 22.078173715 10.3.202.234 142.250.71.174 ICMP 98 Echo (ping) request id= 892 23.102268947 10.3.202.234 142.250.71.174 ICMP 98 Echo (ping) request id= 2214 57.553161339 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2223 57.553234130 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2226 57.553249984 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2231 57.553265980 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2236 57.553298539 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2239 57.553311416 10.3.202.234 142.250.197.227 ICMP 590 Destination unreachable 2252 57.559644075 10.3.202.234 142.250.197.196 ICMP 192 Destination unreachable 2405 58.352898342 10.3.202.234 10.3.200.1 ICMP 370 Destination unreachable 2408 58.364703727 10.3.202.234 10.3.200.1 ICMP 370 Destination unreachable 4090 70.933922154 10.3.202.234 10.3.200.1 ICMP 370 Echo (ping) request id= 4099 70.933940389 10.3.202.234 10.3.200.1 ICMP 370 Destination unreachable 4713 77.051972670 10.3.200.1 10.3.202.234 ICMP 562 Echo (ping) reply id= 6293 84.345712579 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6296 84.354041706 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= 6406 85.347179984 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6410 85.369051853 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= 6677 86.348741703 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6682 86.357349398 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= 6748 87.349929049 10.3.202.234 10.3.200.1 ICMP 882 Echo (ping) request id= 6753 87.362775437 10.3.200.1 10.3.202.234 ICMP 882 Echo (ping) reply id= </pre>
<pre> Frame 6290: 882 bytes on wire (7056 bits), 882 bytes captured (7056 bits) on interface wlan0, id 0 Ethernet II, Src: Cisco 8a:90:e2 (08:4f:a9:8a:90:e2), Dst: AzureWaveTec cc:05:a1 (50:5a:65:cc:05:a1) Internet Protocol Version 4, Src: 10.3.200.1, Dst: 10.3.202.234 0100 = Version: 4 ... 0101 = Header Length: 20 bytes (5) Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 868 Identification: 0x44f6 (17054) 000. = Flags: 0x0 ... 0 0001 0111 0010 = Fragment Offset: 2960 Time to Live: 64 Protocol: ICMP (1) Header Checksum: 0xc3e [validation disabled] [Header checksum status: Unverified] Source Address: 10.3.200.1 Destination Address: 10.3.202.234 * [3 IPv4 Fragments (3888 bytes): #6294(1480), #6295(1480), #6296(948)] [Frame 6294, payload: 0:1478 (1480 bytes)] [Frame 6295, payload: 1480:2959 (1480 bytes)] [Frame 6296, payload: 2960:3007 (848 bytes)] [Fragment count: 3] [Reassembled IPv4 length: 3888] [Reassembled IPv4 data [truncated]: 00002123bc590001a494f8600000000e28a600000000001011213141 Internet Control Message Protocol </pre>	<pre> Frame 6292: 882 bytes on wire (7056 bits), 882 bytes captured (7056 bits) on interface wlan0, id 0 Ethernet II, Src: AzureWaveTec cc:05:a1 (50:5a:65:cc:05:a1), Dst: Cisco 8a:90:e2 (08:4f:a9:8a:90:e2) Internet Protocol Version 4, Src: 10.3.202.234, Dst: 10.3.200.1 0100 = Version: 4 ... 0101 = Header Length: 20 bytes (5) Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 868 Identification: 0x44f6 (17054) 000. = Flags: 0x0 ... 0 0001 0111 0010 = Fragment Offset: 2960 Time to Live: 64 Protocol: ICMP (1) Header Checksum: 0xba3f [validation disabled] [Header checksum status: Unverified] Source Address: 10.3.202.234 Destination Address: 10.3.200.1 * [3 IPv4 Fragments (3888 bytes): #6291(1480), #6292(1480), #6293(948)] [Frame 6291, payload: 0:1478 (1480 bytes)] [Frame 6292, payload: 1480:2959 (1480 bytes)] [Frame 6293, payload: 2960:3007 (848 bytes)] [Fragment count: 3] [Reassembled IPv4 length: 3888] [Reassembled IPv4 data [truncated]: 00001923bc590001a494f8600000000e28a600000000001011213141 Internet Control Message Protocol </pre>

9. Tracert командын үр дүнгээс эхний 2 рүүтэртэй солилцсон датаграмуудыг ажиглая. TTL-ийн утга ямар байна вэ? Өөрчлөгдөж байна уу? TTL exceeded ямар утгыг илэрхийлж байна вэ?

Traceroute нь TTL талбарыг ашиглан зам мөрийг олдог.

- Эхний пакет TTL=1 → 1-р чиглүүлэгч TTL=0 болж ICMP Time Exceeded хариу илгээнэ.
- Дараагийн пакет TTL=2 → 2-р чиглүүлэгч дээр TTL=0 болж ICMP Time Exceeded буцаана.
→ Traceroute энэ хариуг ашиглан зам дахь чиглүүлэгч бүрийн IP-г тодорхойлдог.

TTL утга **өөрчлөгдөж байна**, учир нь traceroute алхам бүрд TTL утга 1-ээр өсдөг. **TTL Exceeded** гэдэг нь тухайн чиглүүлэгч пакетын амьдрах хугацаа дууссаныг илэрхийлдэг бөгөөд пакет устгагдаж, ICMP алдааны мэдэгдэл илгээгддэг.

4690	76.933940389	10.3.202.234	10.3.200.1	ICMP	370 Destination unreachable
4713	77.051972670	10.3.200.1	10.3.202.234	ICMP	562 Echo (ping) reply id=
6293	84.345712579	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6296	84.354641706	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6406	85.347179984	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6410	85.360051853	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6677	86.348741703	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6682	86.357349398	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6748	87.349929049	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6753	87.362775437	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
7664	112.763586041	10.3.200.1	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7666	112.773345909	10.3.200.1	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7668	112.780166229	10.3.200.1	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7670	112.786457496	10.9.1.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7672	112.896778632	10.9.1.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7674	112.902894861	10.9.1.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7676	112.909446867	10.0.2.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7678	112.915964630	10.0.2.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7680	112.922544824	10.0.2.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7682	112.925766930	119.40.100.41	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7684	112.932247883	119.40.100.41	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7686	112.938788741	119.40.100.41	10.3.202.234	ICMP	70 Time-to-live exceeded (T
<pre> > Frame 7668: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlo1, id 0 > Ethernet II, Src: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2), Dst: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1) > Internet Protocol Version 4, Src: 10.3.200.1, Dst: 10.3.202.234 0100 = Version: 4 0101 = Header Length: 20 bytes (5) > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT) Total Length: 56 Identification: 0x4d36 (19766) > 000. = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 254 Protocol: ICMP (1) Header Checksum: 0xc7dc [validation disabled] [Header checksum status: Unverified] Source Address: 10.3.200.1 Destination Address: 10.3.202.234 > Internet Control Message Protocol </pre>					

Эхний router

4690	76.933940389	10.3.202.234	10.3.200.1	ICMP	370 Destination unreachable
4713	77.051972670	10.3.200.1	10.3.202.234	ICMP	562 Echo (ping) reply id=
6293	84.345712579	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6296	84.354641706	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6406	85.347179984	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6410	85.360051853	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6677	86.348741703	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6682	86.357349398	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6748	87.349929049	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6753	87.362775437	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
7664	112.763586041	10.3.200.1	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7666	112.773345909	10.3.200.1	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7668	112.780166229	10.3.200.1	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7670	112.786457496	10.9.1.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7672	112.896778632	10.9.1.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7674	112.902894861	10.9.1.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7676	112.909446867	10.0.2.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7678	112.915964630	10.0.2.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7680	112.922544824	10.0.2.9	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7682	112.925766930	119.40.100.41	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7684	112.932247883	119.40.100.41	10.3.202.234	ICMP	70 Time-to-live exceeded (T
7686	112.938788741	119.40.100.41	10.3.202.234	ICMP	70 Time-to-live exceeded (T
<pre> > Frame 7670: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlo1, id 0 > Ethernet II, Src: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2), Dst: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1) > Internet Protocol Version 4, Src: 10.9.1.9, Dst: 10.3.202.234 0100 = Version: 4 0101 = Header Length: 20 bytes (5) > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT) Total Length: 56 Identification: 0x4fbf (20415) > 000. = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 253 Protocol: ICMP (1) Header Checksum: 0x8d46 [validation disabled] [Header checksum status: Unverified] Source Address: 10.9.1.9 Destination Address: 10.3.202.234 > Internet Control Message Protocol </pre>					

2 дох router

10. Аль Ping командыг гүйцэтгэхэд Fragment хийгдсэн байна бэ? Яагаад?

Ping -s 1536, ping -s 2000 болон ping -s 3800 командууд гүйцэтгэхэд fragmentation үүссэн. Учир нь **MTU (Maximum Transmission Unit)** ихэнх Ethernet сүлжээнд 1500 байт байдаг тул илүү хэмжээтэй датаграмуудыг IP автоматаар хуваадаг.

756	15.178308292	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=
782	15.747241848	10.3.202.234	142.250.71.174	ICMP	554	Echo (ping) request	id=
783	16.069594157	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=
784	16.748706539	10.3.202.234	142.250.71.174	ICMP	554	Echo (ping) request	id=
785	16.862167293	142.250.71.174	10.3.202.234	ICMP	554	Echo (ping) reply	id=
786	16.869485491	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=
787	17.015309240	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=
788	17.871683158	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=
790	18.083929534	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=
798	18.873253583	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=
799	18.989274319	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=
821	19.874681369	10.3.202.234	142.250.71.174	ICMP	1066	Echo (ping) request	id=
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066	Echo (ping) reply	id=
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98	Echo (ping) request	id=
854	21.054376853	10.3.202.234	142.250.71.174	ICMP	98	Echo (ping) request	id=
898	22.078173715	10.3.202.234	142.250.71.174	ICMP	98	Echo (ping) request	id=
932	23.102250947	10.3.202.234	142.250.71.174	ICMP	98	Echo (ping) request	id=
2214	57.553161339	10.3.202.234	142.250.197.227	ICMP	590	Destination unreachable	
2223	57.553234130	10.3.202.234	142.250.197.227	ICMP	590	Destination unreachable	
2226	57.553249984	10.3.202.234	142.250.197.227	ICMP	590	Destination unreachable	
2231	57.553265980	10.3.202.234	142.250.197.227	ICMP	590	Destination unreachable	
2236	57.553298539	10.3.202.234	142.250.197.227	ICMP	590	Destination unreachable	
.....							
▶ Frame 932: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlo1, id 0							
▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)							
▼ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174							
0100 = Version: 4							
.... 0101 = Header Length: 20 bytes (5)							
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)							
Total Length: 84							
Identification: 0x39aa (14762)							
▶ 0000 = Flags: 0x0							
...0 0000 1011 1001 = Fragment Offset: 1480							
Time to Live: 64							
Protocol: ICMP (1)							
Header Checksum: 0x94b0 [validation disabled]							
[Header checksum status: Unverified]							
Source Address: 10.3.202.234							
Destination Address: 142.250.71.174							
▼ [2 IPv4 Fragments (1544 bytes): #931(1480), #932(64)]							
[Frame: 931, payload: 0-1479 (1480 bytes)]							
[Frame: 932, payload: 1480-1543 (64 bytes)]							
[Fragment count: 2]							
[Reassembled IPv4 length: 1544]							
[Reassembled IPv4 data [truncated]: 0800b64bbaa000046794f8680000000dbd30200000000010111213141							
▶ Internet Control Message Protocol							

11. Хуваагдсан датаграмуудаас эхний fragment-ийг харуул. IP header-ийн аль талбар Fragment хийгдсэнийг илэрхийлэж байна вэ?

Wireshark-д эхний fragment-ийн IP header-д:

- **Flags: More Fragments = 1**
- **Fragment Offset = 0**

гэсэн утга гарч ирнэ.
→ Энэ хоёр талбар (Flags ба Fragment Offset) нь датаграм fragmentation хийгдсэн болохыг илтгэнэ.

825	19.993758553	10.3.202.39	255.255.255.255	UDP	100 48689 → 48689 Len=58
826	19.993758650	10.3.201.66	224.0.0.251	MDNS	509 Standard query response
827	19.997399877	10.3.201.71	224.0.0.251	MDNS	412 Standard query response
828	19.997400569	fe80::1480:2804:7bc...	ff02::fb	MDNS	467 Standard query response
829	20.001404638	10.3.201.105	224.0.0.251	MDNS	447 Standard query response
830	20.001405419	10.3.201.83	224.0.0.251	MDNS	437 Standard query response
831	20.005356625	10.3.203.233	224.0.0.251	MDNS	403 Standard query response
832	20.005357359	fe80::1ca6:c587:29c...	ff02::fb	MDNS	423 Standard query response
833	20.005357447	fe80::cdd:8037:fd53...	ff02::fb	MDNS	120 Standard query 0x0000 PT
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply id=
835	20.010701228	10.3.202.234	142.250.71.174	IPv4	1514 Fragmented IP protocol (
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request id=
837	20.909006031	44.205.96.163	10.3.202.234	TLSv1.2	112 Application Data
838	20.909006783	44.205.96.163	10.3.202.234	TLSv1.2	97 Encrypted Alert
839	20.909006907	44.205.96.163	10.3.202.234	TCP	66 443 → 56038 [FIN, ACK] S
840	20.909007016	52.108.44.3	10.3.202.234	TLSv1.2	98 Application Data
841	20.909115557	10.3.202.234	44.205.96.163	TCP	66 [TCP Previous segment no
842	20.909147841	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [ACK] Seq=2
843	20.909789093	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [FIN, ACK] S
844	20.910393143	10.3.202.234	52.108.44.3	TLSv1.2	89 Application Data
845	20.913279926	10.3.202.121	224.0.0.251	MDNS	253 Standard query 0x0000 PT
846	20.913867007	10.3.202.39	224.0.0.251	MDNS	450 Standard query response

▶ Frame 835: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface wlo1, id 6
 ▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)
 ▶ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 1500
 Identification: 0x324d (12877)
 001. = Flags: 0x1, More fragments
 0... = Reserved bit: Not set
 .0... = Don't fragment: Not set
 ..1. = More fragments: Set
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 64
 Protocol: ICMP (1)
 Header Checksum: 0x773e [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 10.3.202.234
 Destination Address: 142.250.71.174
 [Reassembled IPv4 in frame: 836]
 ▶ Data (1480 bytes)

12. Бүх хуваагдсан хэсгүүдийг ажигла. Хэрхэн датаграмын хуваагдлыг эхний ба сүүлийнх гэдгийг тодорхойлж байна вэ?

- **Эхний fragment:** Fragment Offset = 0, More Fragments = 1
- **Сүүлийн fragment:** More Fragments = 0
 → Энэ байдлаар IP нь хуваагдсан датаграмуудын дараалал болон төгсгөлийг тодорхойлдог.

825	19.993758553	10.3.202.39	255.255.255.255	UDP	100 48689 → 48689 Len=58
826	19.993758650	10.3.201.66	224.0.0.251	MDNS	509 Standard query response
827	19.997399877	10.3.201.71	224.0.0.251	MDNS	412 Standard query response
828	19.997400569	fe80::1480:2804:7bc...	ff02::fb	MDNS	467 Standard query response
829	20.001404638	10.3.201.105	224.0.0.251	MDNS	447 Standard query response
830	20.001405419	10.3.201.83	224.0.0.251	MDNS	437 Standard query response
831	20.005356625	10.3.203.233	224.0.0.251	MDNS	403 Standard query response
832	20.005357359	fe80::1ca6:c587:29c...	ff02::fb	MDNS	423 Standard query response
833	20.005357447	fe80::cdd:8037:fd53...	ff02::fb	MDNS	120 Standard query 0x0000 PT
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply id=
835	20.010701228	10.3.202.234	142.250.71.174	IPv4	1514 Fragmented IP protocol (
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request id=
837	20.909006031	44.205.96.163	10.3.202.234	TLSv1.2	112 Application Data
838	20.909006783	44.205.96.163	10.3.202.234	TLSv1.2	97 Encrypted Alert
839	20.909006907	44.205.96.163	10.3.202.234	TCP	66 443 → 56038 [FIN, ACK] S
840	20.909007016	52.108.44.3	10.3.202.234	TLSv1.2	98 Application Data
841	20.909115557	10.3.202.234	44.205.96.163	TCP	66 [TCP Previous segment no
842	20.909147841	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [ACK] Seq=2
843	20.909789093	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [FIN, ACK] S
844	20.910393143	10.3.202.234	52.108.44.3	TLSv1.2	89 Application Data
845	20.913279926	10.3.202.121	224.0.0.251	MDNS	253 Standard query 0x0000 PT
846	20.913867007	10.3.202.39	224.0.0.251	MDNS	450 Standard query response

▶ Frame 836: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlo1, id 0
 ▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)
 ▶ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 84
 Identification: 0x324d (12877)
 ▼ 0000 = Flags: 0x0
 0... = Reserved bit: Not set
 .0... = Don't fragment: Not set
 ..0. = More fragments: Not set
 ...0 0000 1011 1001 = Fragment Offset: 1480
 Time to Live: 64
 Protocol: ICMP (1)
 Header Checksum: 0x9c0d [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 10.3.202.234
 Destination Address: 142.250.71.174
 ▼ [2 IPv4 Fragments (1544 bytes): #835(1480), #836(64)]
 [Frame: 835, payload: 0-1479 (1480 bytes)]
 [Frame: 836, payload: 1480-1543 (64 bytes)]
 [Fragment count: 2]
 [Reassembled IPv4 length: 1544]
 [Reassembled IPv4 data [truncated]: 080022b4baa000016494f86800000000736e01000000000010111213141

825	19.993758553	10.3.202.39	255.255.255.255	UDP	100 48689 → 48689 Len=58
826	19.993758650	10.3.201.66	224.0.0.251	MDNS	509 Standard query response
827	19.997399877	10.3.201.71	224.0.0.251	MDNS	412 Standard query response
828	19.997400569	fe80::1480:2804:7bc...	ff02::fb	MDNS	467 Standard query response
829	20.001404638	10.3.201.105	224.0.0.251	MDNS	447 Standard query response
830	20.001405419	10.3.201.83	224.0.0.251	MDNS	437 Standard query response
831	20.005356625	10.3.203.233	224.0.0.251	MDNS	403 Standard query response
832	20.005357359	fe80::1ca6:c587:29c...	ff02::fb	MDNS	423 Standard query response
833	20.005357447	fe80::cdd:8037:fd53...	ff02::fb	MDNS	120 Standard query 0x0000 PT
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply id=
835	20.010701228	10.3.202.234	142.250.71.174	IPv4	1514 Fragmented IP protocol (
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request id=
837	20.909006031	44.205.96.163	10.3.202.234	TLSv1.2	112 Application Data
838	20.909006783	44.205.96.163	10.3.202.234	TLSv1.2	97 Encrypted Alert
839	20.909006907	44.205.96.163	10.3.202.234	TCP	66 443 → 56038 [FIN, ACK] S
840	20.909007016	52.108.44.3	10.3.202.234	TLSv1.2	98 Application Data
841	20.909115557	10.3.202.234	44.205.96.163	TCP	66 [TCP Previous segment no
842	20.909147841	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [ACK] Seq=2
843	20.909789093	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [FIN, ACK] S
844	20.910393143	10.3.202.234	52.108.44.3	TLSv1.2	89 Application Data
845	20.913279926	10.3.202.121	224.0.0.251	MDNS	253 Standard query 0x0000 PT
846	20.913867007	10.3.202.39	224.0.0.251	MDNS	450 Standard query response


```

▶ Frame 835: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface wlo1, id 6
▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)
▼ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1500
    Identification: 0x324d (12877)
  ▼ 001. .... = Flags: 0x1, More fragments
    0... .... = Reserved bit: Not set
    .0... .... = Don't fragment: Not set
    ..1. .... = More fragments: Set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0x773e [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.3.202.234
    Destination Address: 142.250.71.174
    [Reassembled IPv4 in frame: 836]
  ▶ Data (1480 bytes)

```

Frame 835, 836

13. Эхний болон 2 дахь хуваагдсан датаграмын толгой хэсгийн аль талбарууд өөрчлөгдсөн байна вэ? Тайлбарла.

Өөрчлөгдсөн талбарууд:

- **Total Length:** Хэсэг бүрийн хэмжээ өөр.
- **Fragment Offset:** Дараагийн фрагмент бүрийн байрлалыг заана.
- **Flags:** MF (More Fragments) утга ялгаатай.
- **Header Checksum:** Толгой өөрчлөгдсөн тул дахин тооцогддог. Бусад (Source, Destination, Protocol, Identification) талбарууд ижил.

825	19.993758553	10.3.202.39	255.255.255.255	UDP	100 48689 → 48689 Len=58
826	19.993758650	10.3.201.66	224.0.0.251	MDNS	509 Standard query response
827	19.997399877	10.3.201.71	224.0.0.251	MDNS	412 Standard query response
828	19.997400569	fe80::1480:2804:7bc...	ff02::fb	MDNS	467 Standard query response
829	20.001404638	10.3.201.105	224.0.0.251	MDNS	447 Standard query response
830	20.001405419	10.3.201.83	224.0.0.251	MDNS	437 Standard query response
831	20.005356625	10.3.203.233	224.0.0.251	MDNS	403 Standard query response
832	20.005357359	fe80::1ca6:c587:29c...	ff02::fb	MDNS	423 Standard query response
833	20.005357447	fe80::cdd:8037:fd53...	ff02::fb	MDNS	120 Standard query 0x0000 PT
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply id=
835	20.010701228	10.3.202.234	142.250.71.174	IPv4	1514 Fragmented IP protocol (
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request id=
837	20.909006031	44.205.96.163	10.3.202.234	TLSv1.2	112 Application Data
838	20.909006783	44.205.96.163	10.3.202.234	TLSv1.2	97 Encrypted Alert
839	20.909006907	44.205.96.163	10.3.202.234	TCP	66 443 → 56038 [FIN, ACK] S
840	20.909007016	52.108.44.3	10.3.202.234	TLSv1.2	98 Application Data
841	20.909115557	10.3.202.234	44.205.96.163	TCP	66 [TCP Previous segment no
842	20.909147841	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [ACK] Seq=2
843	20.909789093	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [FIN, ACK] S
844	20.910393143	10.3.202.234	52.108.44.3	TLSv1.2	89 Application Data
845	20.913279926	10.3.202.121	224.0.0.251	MDNS	253 Standard query 0x0000 PT
846	20.913867007	10.3.202.39	224.0.0.251	MDNS	450 Standard query response

▶ Frame 836: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlo1, id 0
 ▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)
 ▶ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 84
 Identification: 0x324d (12877)
 ▼ 0000 = Flags: 0x0
 0... = Reserved bit: Not set
 .0... = Don't fragment: Not set
 ..0. = More fragments: Not set
 ...0 0000 1011 1001 = Fragment Offset: 1480
 Time to Live: 64
 Protocol: ICMP (1)
 Header Checksum: 0x9c0d [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 10.3.202.234
 Destination Address: 142.250.71.174
 ▼ [2 IPv4 Fragments (1544 bytes): #835(1480), #836(64)]
 [Frame: 835, payload: 0-1479 (1480 bytes)]
 [Frame: 836, payload: 1480-1543 (64 bytes)]
 [Fragment count: 2]
 [Reassembled IPv4 length: 1544]
 [Reassembled IPv4 data [truncated]: 080022b4baa000016494f86800000000736e01000000000010111213141

825	19.993758553	10.3.202.39	255.255.255.255	UDP	100 48689 → 48689 Len=58
826	19.993758650	10.3.201.66	224.0.0.251	MDNS	509 Standard query response
827	19.997399877	10.3.201.71	224.0.0.251	MDNS	412 Standard query response
828	19.997400569	fe80::1480:2804:7bc...	ff02::fb	MDNS	467 Standard query response
829	20.001404638	10.3.201.105	224.0.0.251	MDNS	447 Standard query response
830	20.001405419	10.3.201.83	224.0.0.251	MDNS	437 Standard query response
831	20.005356625	10.3.203.233	224.0.0.251	MDNS	403 Standard query response
832	20.005357359	fe80::1ca6:c587:29c...	ff02::fb	MDNS	423 Standard query response
833	20.005357447	fe80::cdd:8037:fd53...	ff02::fb	MDNS	120 Standard query 0x0000 PT
834	20.005357528	142.250.71.174	10.3.202.234	ICMP	1066 Echo (ping) reply id=
835	20.010701228	10.3.202.234	142.250.71.174	IPv4	1514 Fragmented IP protocol (
836	20.010724355	10.3.202.234	142.250.71.174	ICMP	98 Echo (ping) request id=
837	20.909006031	44.205.96.163	10.3.202.234	TLSv1.2	112 Application Data
838	20.909006783	44.205.96.163	10.3.202.234	TLSv1.2	97 Encrypted Alert
839	20.909006907	44.205.96.163	10.3.202.234	TCP	66 443 → 56038 [FIN, ACK] S
840	20.909007016	52.108.44.3	10.3.202.234	TLSv1.2	98 Application Data
841	20.909115557	10.3.202.234	44.205.96.163	TCP	66 [TCP Previous segment no
842	20.909147841	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [ACK] Seq=2
843	20.909789093	10.3.202.234	44.205.96.163	TCP	66 56038 → 443 [FIN, ACK] S
844	20.910393143	10.3.202.234	52.108.44.3	TLSv1.2	89 Application Data
845	20.913279926	10.3.202.121	224.0.0.251	MDNS	253 Standard query 0x0000 PT
846	20.913867007	10.3.202.39	224.0.0.251	MDNS	450 Standard query response

```

▶ Frame 835: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface wlo1, id 6
▶ Ethernet II, Src: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1), Dst: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2)
▼ Internet Protocol Version 4, Src: 10.3.202.234, Dst: 142.250.71.174
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1500
    Identification: 0x324d (12877)
  ▼ 001. .... = Flags: 0x1, More fragments
    0... .... = Reserved bit: Not set
    .0... .... = Don't fragment: Not set
    ..1. .... = More fragments: Set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0x773e [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.3.202.234
    Destination Address: 142.250.71.174
    [Reassembled IPv4 in frame: 836]
  ▶ Data (1480 bytes)

```

14. Датаграмыг хуваахдаа хэмжээг хэрхэн тогтоох вэ? Хуваагдсан датаграмууд дээр тайлбарла.

IP датаграмыг хуваахдаа **MTU (Maximum Transmission Unit)**-ийн хэмжээнд тулгуурладаг. Ethernet сүлжээнд MTU = 1500 bytes байдаг. Тиймээс:

- IP header (20 bytes)-ийг хасаж
- Үлдсэн 1480 байт бүрийг payload болгон илгээдэг. Хэрэв 3800 байт өгөгдөлтэй пакет илгээвэл дараах байдлаар хуваагдана:
 1. 1480 байт payload
 2. 1480 байт payload
 3. 840 байт payload + (сүүлийн хэсэг, MF=0)

6742	87.222659934	10.3.202.234	142.250.71.229	TCP	1466 48820 → 443 [ACK] Seq=42
6743	87.222678788	10.3.202.234	142.250.71.229	TCP	1466 48820 → 443 [PSH, ACK] S
6744	87.222689813	10.3.202.234	142.250.71.229	TLSv1.2	1331 Application Data
6745	87.222898809	10.3.202.234	10.0.80.8	DNS	86 Standard query 0x9961 A
6746	87.349870565	10.3.202.234	10.3.200.1	IPv4	1514 Fragmented IP protocol (
6747	87.349924006	10.3.202.234	10.3.200.1	IPv4	1514 Fragmented IP protocol (
6748	87.349929049	10.3.202.234	10.3.200.1	ICMP	882 Echo (ping) request id=
6749	87.353098801	10.3.202.234	142.250.71.229	TLSv1.2	215 Application Data, Applic
6750	87.359569065	10.0.80.8	10.3.202.234	DNS	102 Standard query response
6751	87.362774835	10.3.200.1	10.3.202.234	IPv4	1514 Fragmented IP protocol (
6752	87.362775338	10.3.200.1	10.3.202.234	IPv4	1514 Fragmented IP protocol (
6753	87.362775437	10.3.200.1	10.3.202.234	ICMP	882 Echo (ping) reply id=
6754	87.530214832	142.250.71.229	10.3.202.234	TCP	66 443 → 48820 [ACK] Seq=16
6755	87.530215711	142.250.71.229	10.3.202.234	TCP	66 443 → 48820 [ACK] Seq=16
6756	87.530215874	142.250.71.229	10.3.202.234	TCP	66 443 → 48820 [ACK] Seq=16
6757	87.530216021	142.250.71.229	10.3.202.234	TCP	66 443 → 48820 [ACK] Seq=16
6758	87.530216164	142.250.71.229	10.3.202.234	TLSv1.2	105 Application Data
6759	87.571075620	10.3.202.234	142.250.71.229	TCP	66 48820 → 443 [ACK] Seq=85
6760	87.709679988	142.250.71.229	10.3.202.234	TLSv1.2	1056 Application Data
6761	87.709681018	142.250.71.229	10.3.202.234	TLSv1.2	421 Application Data
6762	87.709681159	142.250.71.229	10.3.202.234	TLSv1.2	229 Application Data
6763	87.709681282	142.250.71.229	10.3.202.234	TLSv1.2	105 Application Data


```

▶ Frame 6753: 882 bytes on wire (7056 bits), 882 bytes captured (7056 bits) on interface wlo1, id 0
▶ Ethernet II, Src: Cisco_6a:90:e2 (08:4f:a9:6a:90:e2), Dst: AzureWaveTec_cc:65:a1 (50:5a:65:cc:65:a1)
▼ Internet Protocol Version 4, Src: 10.3.200.1, Dst: 10.3.202.234
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 868
    Identification: 0x4adc (19164)
  ▼ 0000 .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0... .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0001 0111 0010 = Fragment Offset: 2960
    Time to Live: 254
    Protocol: ICMP (1)
    Header Checksum: 0xc658 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.3.200.1
    Destination Address: 10.3.202.234
  ▼ [3 IPv4 Fragments (3808 bytes): #6751(1480), #6752(1480), #6753(848)]
    [Frame: 6751, payload: 0-1479 (1480 bytes)]
    [Frame: 6752, payload: 1480-2959 (1480 bytes)]
    [Frame: 6753, payload: 2960-3807 (848 bytes)]
    [Fragment count: 3]
    [Reassembled IPv4 length: 3808]

```

5.5 Сорих асуулт

1. TTL-ийн утгыг тайлбарлана уу?

Time To Live — пакет сүлжээгээр дамжих hop-ийн тоог хязгаарлах зориулалттай 8 битийн талбар. TTL=0 болох үед пакет устгагдаж ICMP Time Exceeded илгээгддэг.

2. Identification талбарын утга юуг илэрхийлдэг вэ?

Нэг IP датаграммын бүх фрагментүүдийг ялгах, дахин нэгтгэхэд ашиглагддаг өвөрмөц дугаар.

3. Сүлжээгээр дамжиж буй пакет нь чиглүүлэгч төхөөрөмжөөр дамжин өнгөрөх үед IP header-ийн TTL талбарын утга 1 байв. Чиглүүлэгч төхөөрөмж ямар үйлдэл хийх вэ?

Пакет дамжих үед TTL=0 болно → чиглүүлэгч пакетыг устгаад ICMP Time Exceeded илгээнэ.

4. IP header-ийн TOS (Type of Service) талбар ямар утгыг илэрхийлдэг вэ?

Type of Service — сүлжээний чанарын түвшин, хурд, хоцрогдол, найдвартай байдал зэргийг зааж өгдөг (өнөө үед DSCP талбараар орлуулсан).

5. IP datagram-ийн checksum хэрхэн тооцоолох вэ? Жишээгээр тайлбарла.

IP Header Checksum (Толгойн хяналтын нийлбэр) нь IP датаграмын **зөвхөн толгой хэсэгт** (IP Header) алдаа гарсан эсэхийг шалгадаг 16 битийн талбар юм.

- **Тооцоолох Аргачлал:**

- IP толгой хэсгийг 16 битийн (2 байтын) бүлгүүдэд хуваах.
- Checksum талбарыг **0**-ээр дүүргэх.
- 16 битийн бүлгүүд бүрийн **нэгжийн нэмэгдэл** (One's Complement) ашиглан нийлбэрийг олох.
- Гарааны нийлбэрийн **нэгжийн нэмэгдэл**-ийг авч, түүнийг **Checksum** талбарт бичих.

- **Хүлээн авагчид шалгах:** Хүлээн авагч пакетыг хүлээн авч, толгой хэсгийн бүх 16 битийн бүлгүүдийг (Checksum-ийг оруулаад) нэмээд, **нэгжийн нэмэгдэл**-ийг авч нийлбэрийг олох. Хэрэв үр дүн **0** байвал толгой хэсэг **алдаагүй** гэж тооцдог.

Жишээ:

- IP толгойн хэсгийн эхний хоёр 16 битийн утга 0x4500 ба 0x003C байг. Checksum талбар 0x0000 байг.
- Нэмэх: $0x4500 + 0x003C = 0x453C$.
- Толгой хэсгийн бүх 16 битийн нийлбэр S гарсан гэе.
- **Checksum** = S-ийн **нэгжийн нэмэгдэл** (One's Complement).